openSUSE Asia Summit 2026
Yogyakarta, October 3 - 4

Container Observability on openSUSE

A Practical Guide with Prometheus, cAdvisor, and Grafana

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From Ubuntu habits to openSUSE Tumbleweed

- **My day job:** running infrastructure on Ubuntu-based servers
- Most container monitoring guides assume apt, ufw and AppArmor
- **This talk:** the same monitoring stack rebuilt on openSUSE Tumbleweed, with every difference noted
- **My test setup:** Tumbleweed minimal install on a cloud server with a public IP, SELinux enforcing

Agenda

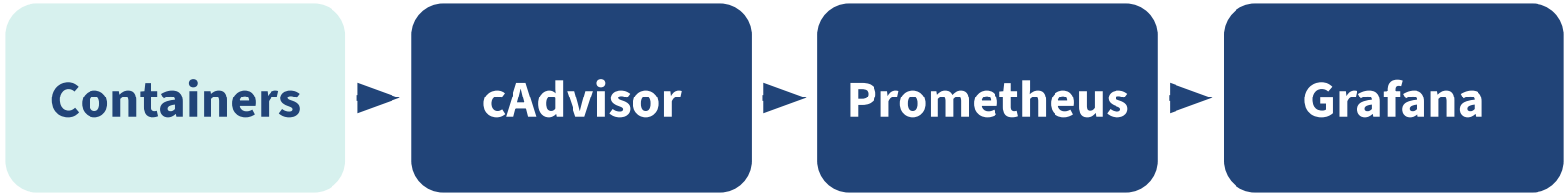
1. Why container observability
2. The stack: cAdvisor, Prometheus, Grafana
3. Installing Docker with zypper
4. Running cAdvisor
5. Configuring Prometheus scraping
6. Building Grafana dashboards
7. openSUSE vs Ubuntu: what changed

Why Container Observability

docker stats shows the present, not the history

- A restarted container takes its **in-memory numbers** with it
- A container can hit its memory limit and get **OOM-killed** while the host looks healthy
- We need **per-container CPU, memory, network and disk**, stored over time and queryable

Three tools, one pipeline: collect, store, visualize



Your workloads, one cgroup each

Reads cgroups, serves /metrics on :8080

Scrapes every 15 s and stores time series on :9090

Runs PromQL and draws dashboards on :3000

Prometheus pulls from cAdvisor; Grafana queries Prometheus. Both are plain HTTP.

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Building the Stack on Tumbleweed

Docker installs straight from the openSUSE repositories

```
sudo zypper refresh
sudo zypper install docker docker-compose
sudo systemctl enable --now docker
sudo usermod -aG docker $USER   # log out and in again

docker compose version
docker info | grep -A1 "Storage Driver"
```

No extra Docker repository or GPG key, unlike the docker-ce setup most Ubuntu guides use.

cAdvisor reads container stats from read-only host mounts

```
cadvisor:
  image: ghcr.io/google/cadvisor:v0.60.5  # moved from gcr.io
  privileged: true
  devices: ["/dev/kmsg"]
  volumes:
    - /:/rootfs:ro
    - /var/run/docker.sock:/var/run/docker.sock:ro
    - type: bind  # Docker-managed containerd on Tumbleweed
      source: /run/docker/containerd/containerd.sock
      target: /run/containerd/containerd.sock
      read_only: true
    - /sys:/sys:ro
    - /var/lib/docker/:/var/lib/docker:ro
    - /dev/disk/:/dev/disk:ro
```

Prometheus scrapes cAdvisor every 15 seconds

```
# prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: cadvisor
    static_configs:
      - targets: ["cadvisor:8080"]
```

Before building dashboards, open the Prometheus Targets page and **confirm the cadvisor job is UP**.

Grafana reaches Prometheus by its service name

```
prometheus:
  image: prom/prometheus
  volumes:   # :Z relabels the file for SELinux
    - ./prometheus.yml:/etc/prometheus/prometheus.yml:ro,Z
  ports: ["127.0.0.1:9090:9090"]
grafana:
  image: grafana/grafana
  ports: ["3000:3000"]
  volumes: ["grafana-data:/var/lib/grafana"]
```

In Grafana, add a Prometheus data source at **http://prometheus:9090**, then import **dashboard 14282** as a starting point.

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Dashboards for CPU, Memory, Network and Disk

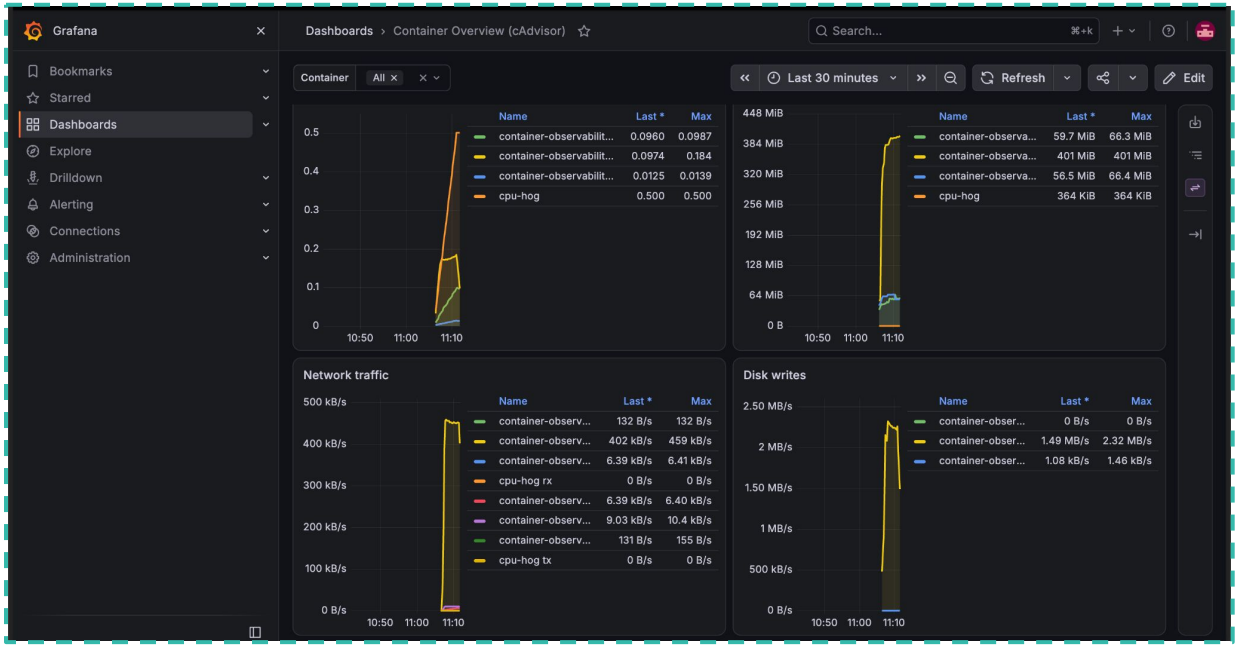
Four PromQL queries cover CPU, memory, network and disk

Panel	PromQL	Unit
CPU	`sum by (name) (rate(container_cpu_usage_seconds_total{name!=""}[5m]))`	cores
Memory	`sum by (name) (container_memory_working_set_bytes{name!=""})`	bytes
Network	`sum by (name) (rate(container_network_receive_bytes_total{name!=""}[5m]))`	bytes/s
Disk	`sum by (name) (rate(container_fs_writes_bytes_total{name!=""}[5m]))`	bytes/s

Use working set memory, not raw usage, for alerts

- **container_memory_usage_bytes** includes page cache the kernel can reclaim
- **container_memory_working_set_bytes** subtracts inactive file cache, so it tracks what leads to an OOM kill
- Filter **name!=""** to drop host-level cgroups such as system.slice
- Add a dashboard variable: **label_values(container_last_seen, name)**

cpu-hog stays at exactly 0.5 cores: the limit at work



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openSUSE vs Ubuntu: What Changed

Most differences are in packaging and security defaults

	Ubuntu (typical guide)	openSUSE Tumbleweed
Packages	apt	zypper
Docker source	Docker's own apt repo (docker-ce)	openSUSE OSS repo (docker)
Firewall	ufw	firewalld
Security module	AppArmor	SELinux, enforcing on new installs[1]
Root filesystem	ext4	Btrfs with Snapper snapshots

[1] Default for new installs since snapshot 20250211. Source: openSUSE News, Feb 2025.

Three checks before you trust the dashboard

SELinux labels

A bind-mounted config fails with permission denied. Add :Z to the volume and read ausearch -m avc. Keep SELinux enforcing.

Storage backend

Docker 29 here uses the containerd image store (overlayfs). cAdvisor then needs the containerd socket, kept in /run/docker/containerd on Tumbleweed. Without it, names vanish.

Published ports

Docker writes its own firewall rules, so a published port can be reachable even when firewalld does not list it. Bind Prometheus to 127.0.0.1 and leave cAdvisor unpublished.

The stack stays the same; openSUSE changes the plumbing

- **zypper** gives you Docker and Compose from the openSUSE repos; the stack itself runs as containers
- Check **SELinux labels, Docker's storage backend and published ports** before you trust the numbers
- Start from Grafana **dashboard 14282**, then add your own PromQL panels
- **Docs:** github.com/google/cadvisor, prometheus.io/docs, grafana.com/docs

Thank You



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Code, config & dashboard

github.com/sintiasnn/container-observability-opensuse

What about Podman? Same stack, a few changes

- Enable the Podman API socket: `sudo systemctl enable --now podman.socket`
- Point cAdvisor at it: `--podman=unix:///var/run/podman/podman.sock`
- Mount `/var/lib/containers` instead of `/var/lib/docker`
- **Start rootful:** rootless needs a per-user socket and still has open cAdvisor issues
- **Not tested for this talk yet:** my next experiment on Tumbleweed